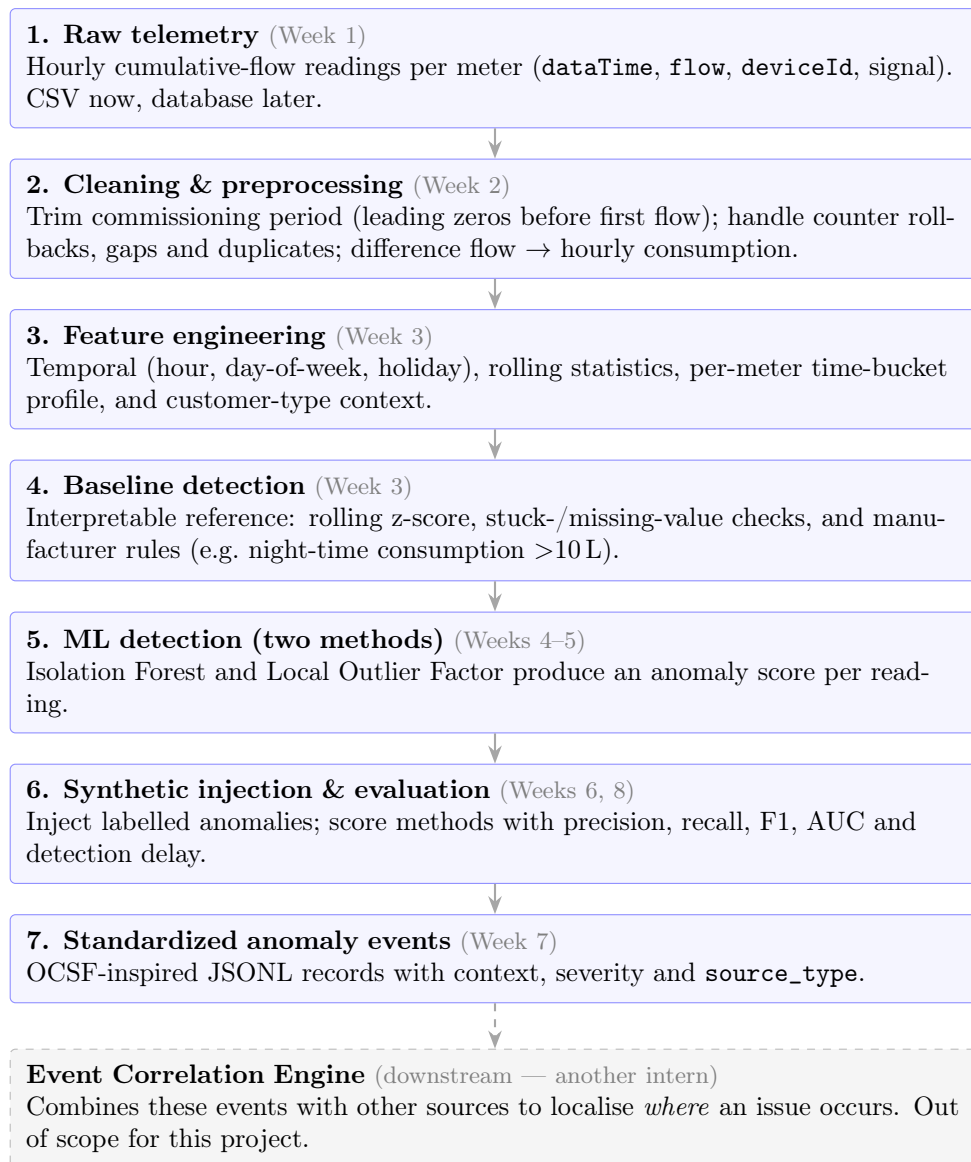


Week 1 Design Note

ML-Based Anomaly and Attack Detection for Smart-Meter Telemetry

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Pipeline Overview



Dataset Description and Access Method

Overview

The dataset consists of NB-IoT smart water-meter telemetry from the operational platform of a Cyprus water utility (EOA), covering **2,031 meters** in the Nicosia district (for now). The

project focuses on the **flow time series**: each meter records a cumulative water-flow counter hourly. Development currently uses

Platform Architecture (context)

The data path is: smart meter → NB-IoT → server firewall → access gateway → application server (a UDP server receives meter packets) → database server. Storage is split across two databases.

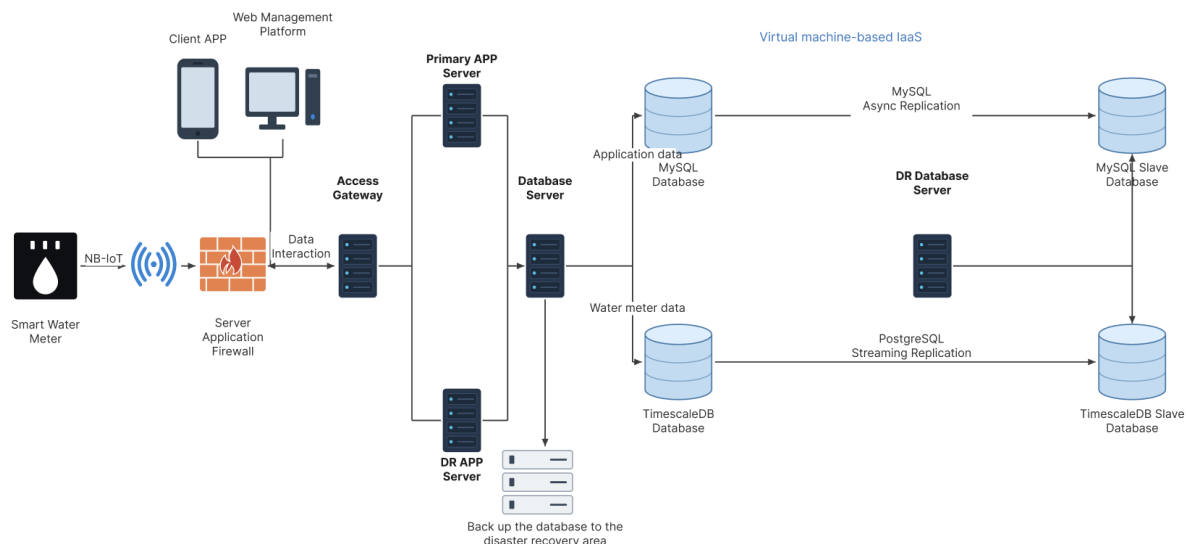


Figure 1: Data Flow Architecture

- **TimescaleDB** (PostgreSQL + time-series extension): the IoT time-series store: raw packets, parsed hourly readings, flow, signal quality, alarms, and daily/monthly aggregates. Uses 7-day chunks, continuous aggregates, and ~1-year retention. **This project's data lives here.**
- **MySQL**: the business store: device profiles, user accounts, billing records, billing/tariff templates, permissions, and DMA analysis. **The customer/building category lives here.**

Two architecture facts matter directly for this project:

- **UDP buffering.** If the application server is unavailable, meter packets are cached as files on the gateway and manually re-synced after recovery. Gaps and delayed/batched arrivals are therefore partly an infrastructure artifact, not necessarily a meter fault. This is relevant to interpreting the “transmission gap” anomaly (Section 2) and to the network-vs-operational distinction.
- **RBAC access.** The platform supports role-based access with `dept_id` data isolation and a built-in *read-only* role. This is the clean, sanctioned mechanism for the data access requested below.

Structure

Two export schemas exist, for the two platform views:

Snapshot export (one row per meter, latest state).

Key fields:

Device SN, **Last communication time**, **Normal Flow (m³)** (cumulative forward-flow

counter), **Back Flow (m³)** (cumulative reverse-flow counter), **Flow Rate (m³/h)** (instantaneous rate), **Alarm** (the platform’s own anomaly flags), **Temperature (°C)**, **Volt (V)**, **Module Battery Voltage (V)**, **RSRP**, **RSSI**, **CSQ** (cellular signal-quality metrics (see Appendix A.1)), **Location / customer fields** (City, Street, Street No, Flat No, Longitude, Latitude, Consumer Number, Device Name, Remarks), **Sleep Times**, **Communication Abnormality Times** (device/network counters).

Historical export (one row per meter per hour — the time series, and the project’s primary input) — 9 fields: **dateTime** (measurement timestamp — the time index), **flow** (cumulative Normal Flow), **deviceId**, **signalCsq**, **isWarning** (warning flag), plus **updatedAt**, **id**, **params**, **paValue** (not needed for analysis).

Historical export is the analysis input and the snapshot is a per-meter context/metadata table joined by **deviceId**.

Sampling and Coverage

- **Sampling interval:** hourly. Meters measure once per hour and transmit the buffered readings to the base station twice per day. Native stored resolution is hourly.
- **Number of meters:** 2,031 in the fleet.
- **Time range:** the 5 obtained meters (for understanding structure - using a web scraper) span ~7–12 months of hourly history each.

Access Method

Decision. The target data-access mechanism is **read-only access to the TimescaleDB time-series store** (via the platform’s RBAC read-only role, scoped by **dept_id**. Why: a repeatable manual extract keeps a person in the loop for every new meter or data refresh. It is not reproducible, does not scale, and breaks at handover. Read-only DB access lets Stage 1 query the source directly: reproducible, scalable to the whole fleet (as it grows), and durable. It also matches the pipeline’s design seam: only Stage 1 changes.

Billing data is separately available, and the MySQL DB store holds the tariff template (customer category) per meter (see Appendix A.4). One time access to this would be great:

Decision. We shall **both** request the meter → tariff-template join from MySQL (personal information can be hidden) *and* build behavioral clustering of meters from their consumption shape. *Why both:* clustering works on the time series we already have, needs no extra access, and is the resilient core. It is also the fallback if the MySQL join never arrives. If the true category is obtained, it serves two purposes: (i) a direct context feature that improves detection, and (ii) ground truth to validate that the clustering recovers the real categories. The two are complementary, not redundant.

Known Limitations

- **Limited annual coverage** — history is insufficient to learn full annual seasonality (a real concern in Cyprus, where summer inflates usage). Daily and weekly patterns are well captured; annual is not.
- **Transmission gaps are partly architectural** — the UDP-buffering / manual-resync mechanism means some gaps reflect infrastructure events, not meter faults.

Anomaly Taxonomy

Anomaly type	Description (in this data)	Category	Possible cause
Sudden spike	Single reading far above local norm	Point	Burst, surge, meter fault
Sudden drop	Single reading far below expected	Point	Supply cut, sensor dropout
Counter rollback	Cumulative flow <i>decreases</i> between consecutive readings	Point	Meter reset/replacement, data error, tampering
Stuck value	Consumption flat (zero diff) over many consecutive hours	Collective	Frozen sensor <i>or</i> genuine zero usage
Measurement gap	Missing hourly reading	Point / Interval	Meter failed to record
Transmission gap	A whole twice-daily batch absent	Collective	Missed transmission / network failure; also UDP-buffering resync (architectural)
Night-time burst	Elevated consumption during typically-quiet hours	Contextual	Leak, tampering (normal for some customer types)
Abnormal profile	Daily/weekly shape deviates from the meter's own normal	Contextual	Behaviour change, leak, misuse
Replay-like pattern	A past window of readings reappears identically	Collective	Data manipulation
Time-shifted profile	Daily profile phase-shifted by N hours	Collective	Clock drift, timezone error, timestamp manipulation
Signal degradation	Sudden sustained drop in RSRP/RSSI	Collective / Contextual	Antenna/hardware fault, environment (network-side anomaly)

Context Enrichment

Context enrichment is what separates “this value is statistically unusual” from “this value is unusual given everything we know about this meter and this moment.” The same raw signal is scored differently depending on context — high usage at 1am is normal for a business but anomalous for a residence. This feeds the feature engineering in Week 3 and improves the scoring of both ML methods.

Temporal Context

Derived directly from each reading's timestamp: **Hour of day, Day of week, Public holiday flag, Season, Time since last reading**

Asset Attributes (Customer Category)

The billing system assigns every meter a tariff template: **Type A (Resident)**, **Type B (Business)**, or **Type C (Industry)**. A school, a warehouse and a home have entirely different “normal” profiles, so category sharpens the anomaly scoring.

Spatial / Hierarchical Context

Master meter, Bulk meter, sub-meter etc. Spatial awareness data, will help our anomaly detection model understand the physical context of water flow. Instead of analyzing each meter independently, the model should compare flows between upstream and downstream meters, detect imbalances, localize leaks or meter faults to specific network segments, and distinguish system-wide demand changes from isolated anomalies. This typically improves both detection accuracy and the ability to identify the likely source and impact area of an issue.

Decision. We should also request **network topology and meter hierarchy** data such as Parent meter, Child meters, Supply zone/DMA (District Metered Area) assignment, Geographic coordinates or GIS location of meters, Pipe/network connectivity between meters and assets, Any existing network schematics or GIS layers describing the distribution system.

Behavioral Baseline Context

Learned from each meter’s own history. The most powerful form of enrichment, because it generalities the per-customer-type idea: each meter establishes its own normal and deviation is measured against that.

- **Per-meter, per-time-bucket profile** — the meter’s typical consumption for this hour-of-day and day-of-week.
- **Rolling statistics** — the meter’s own mean and standard deviation over a recent window.
- **Recent trend** — is consumption gradually rising?

Network-Health Context (Secondary Source)

The cellular metrics (**RSRP**, **RSSI**, **CSQ** and the device counters (**Communication Abnormality Times**, **Sleep Times**) form the network/base-station view. They explain *transmission* problems and let the project separate a network-side fault from an operational (water) anomaly.

How Context Enters Detection

1. **As features** — context such as `hour_of_day`, `customer_category`, and `is_holiday` is added to the feature matrix, and the ML methods learn how it interacts with the value. Primary approach for the ML methods.
2. **As conditioning** — separate expected-profiles per context bucket, with deviation measured against the matching bucket. More interpretable; used for the Week 3 baseline.

Scope note: the per-meter, per-time-bucket baseline plus basic temporal features plus customer category provide most of the value. Spatial topology and weather are higher-effort additions — pursued only after the core works.

Initial Evaluation Plan

Supervision Approach

No reliable ground-truth anomaly labels are available. Detection will therefore be **unsupervised** — methods that learn normal behavior and flag deviations (Isolation Forest and LOF). To enable

quantitative evaluation, **synthetic labels** will be generated in Week 6 via controlled anomaly injection. The platform Alarm flags (leakage, etc.) will additionally be used as a *weak-label* cross-check against real events (if granted access). Where a clearly-normal period can be identified during cleaning, models may be trained primarily on it (a light semi-supervised refinement).

Evaluation Metrics

Methods will be compared using (full definitions and ranges in Appendix A.2):

Precision, Recall, F1-score, AUC-ROC, AUC-PR, Detection delay, Runtime and interpretability

Plain accuracy is deliberately not used: with rare anomalies, a detector that flags nothing scores ~99% while being useless.

Comparison Strategy

All methods — the interpretable baseline, Method 1, and Method 2 — will be evaluated on the same cleaned dataset and on the synthetic anomaly set from Week 6. Results will be presented in a comparison table showing each metric per method, and where possible per anomaly type, so strengths and weaknesses across point/contextual/collective anomalies are visible.

Secondary Validation: Clustering vs. True Categories

If the customer category is obtained from MySQL, it provides a secondary validation distinct from the anomaly metrics above: the behavioral clustering can be checked against the true Type A/B/C labels to confirm it recovers real categories. This validates the context layer, not the anomaly detector itself, but a clustering that aligns with reality strengthens confidence in the context-aware scoring.

Primary Source of Labeled Data

Controlled anomaly injection (Week 6) is the primary mechanism for obtaining labeled evaluation cases. Injection types follow the anomaly taxonomy. Each injected window is recorded with its type, affected meter, start time, duration, and intensity, so detection rate can be measured exactly.

Draft Event Correlation Engine-Compatible Event Schema

Format

Each detected anomaly will be exported as a single JSON record in a `.jsonl` file (one record per line), inspired by the Open Cybersecurity Schema Framework (OCSF) conventions, and agreed jointly with the Event Correlation Engine intern. See field definitions and Example record for the as of now output idea that will be fed to the correlation engine.

Two Telemetry Sources

The project will ideally consider two streams: the **operational view** (flow, consumption — the primary input) and the **network/base-station view** (RSRP, RSSI, CSQ, communication-abnormality counters). Every event carries a `source_type` field so the correlation engine can reason about *where* an issue occurs — e.g. a gap in the operational view that is *explained* by poor signal (or a UDP-buffering resync) in the network view localises the fault to the network layer rather than the water system.

Field Definitions

Field	Type	Description
time	string (ISO 8601)	Measurement timestamp of the detected anomaly
source_type	string	operational_view or network_base_station
asset_id	string	Device SN of the affected meter
metric	string	Measured variable (e.g. hourly_consumption_m3)
observed_value	float	Actual value at the anomaly time
expected_value	float or null	Expected value from the meter's profile, if available
anomaly_score	float [0–1]	Degree of anomaly
method	string	Detection method that produced the event
severity	string	low / medium / high / critical
confidence	float [0–1]	Confidence in the detection
anomaly_type	string	Label from the taxonomy (Section 2)
context	object	Context used in scoring (Section 3)
evidence	string	Human-readable explanation
metadata	object	Location, customer category, data source, etc.

Example Record

```
{
  "time": "2026-07-03T03:00:00Z",
  "source_type": "operational_view",
  "asset_id": "202405101856",
  "metric": "hourly_consumption_m3",
  "observed_value": 2.4,
  "expected_value": 0.0,
  "anomaly_score": 0.87,
  "method": "isolation_forest",
  "severity": "high",
  "confidence": 0.82,
  "anomaly_type": "night_time_burst",
  "context": {
    "customer_category": "Type A (Resident)",
    "hour_of_day": 3,
    "day_of_week": "Friday",
    "is_holiday": false,
    "expected_profile_value": 0.0
  },
  "evidence": "consumption during a normally-zero night hour for a residential meter",
  "metadata": {
    "city": "Nicosia",
    "longitude": 33.147098,
    "latitude": 35.113515,
    "data_source": "EOA"
  }
}
```

Appendix: Reference Information

Reference material gathered during Week 1 — metric definitions, value ranges, platform vocabularies, and architecture notes — for use throughout the project.

A.1 Cellular Network Signal-Quality Metrics

These describe the quality of the meter’s NB-IoT connection to the base station and form the network/base-station source. Weak or degrading signal explains transmission gaps.

Metric	Unit / scale	Range and meaning
RSRP	dBm	Primary LTE/NB-IoT signal-strength metric; isolates the reference signal, filtering noise. > -80 excellent; -80 to -90 good; -90 to -100 fair; -100 to -110 poor; < -110 marginal (practical floor ≈ -140).
RSSI	dBm	Total received power = signal + noise + interference; broadest, least precise of <i>useful</i> signal. More negative = weaker (e.g. -70 strong, -110 weak).
CSQ	integer 0–31 (99 = unknown)	Higher is better. 0–9 marginal/poor; 10–14 ok; 15–19 good; 20–31 excellent. Historically derived from RSSI via the modem AT+CSQ command.

A.2 Evaluation Metrics

Built from the confusion-matrix counts: TP (true positive), FP (false positive), FN (false negative), TN (true negative).

Metric	Definition	Range and meaning
Precision	$TP/(TP + FP)$	0–1. Of flagged anomalies, fraction genuine. Low $\Rightarrow$ false alarms.
Recall (TPR)	$TP/(TP + FN)$	0–1. Of real anomalies, fraction caught. Low $\Rightarrow$ misses.
F1	$2PR/(P + R)$	0–1. Harmonic mean; high only when both precision and recall are high.
AUC-ROC	area under TPR vs FPR	0.5 (useless) to 1.0 (perfect); = probability a random anomaly scores above a random normal. Threshold-independent. Optimistic under heavy imbalance.
AUC-PR	area under precision vs recall	base-rate to 1.0; more honest than ROC when anomalies are rare.
Detection delay	onset $\rightarrow$ first flag	In hours here; lower is better.

Plain accuracy is avoided: with rare anomalies, flagging nothing scores $\sim 99\%$ yet is useless.

A.3 Platform Alarm Vocabulary (weak labels)

The platform raises the following alarms, defined by the meter manufacturer. They originate at three levels — **Dashboard** (platform-side, based on missing communication), **Platform** (computed from consumption rules), and **Meter** (reported by the device itself). These serve as weak labels and as ready-made detection rules to compare against.

Alarm	Source	Definition
Sleeping Device	Dashboard	Data not received by the platform for more than 3 but fewer than 4 days.
Communication Alarm	Dashboard	Data not received by the platform for 4 days or more.
Logic Leakage Alarm	Platform	Water consumption exceeds 10 L in both the 02:00–03:00 and 03:00–04:00 windows.
Reverse Inst.	Platform	The meter shows reverse flow and no positive flow for 10 days.
Leakage Of Device	Meter	Instantaneous flow persists for more than 1800 seconds.
Air Tube	Meter	The pipe is not full of water (air detected in the tube).
Battery Low	Meter	Battery voltage is below 3.3 Vdc.
Water Temperature Low	Meter	Water temperature is below 4 °C.

Use as weak labels for partial validation only — not ground truth. The Dashboard/Meter alarms map to the transmission-gap and device-health anomaly types; the Platform alarms (Logic Leakage, Reverse Inst.) are consumption-rule detections that directly parallel the abnormal-consumption and counter-rollback entries in the taxonomy.

A.4 Data Fields and Device-Health Reference

- **Normal Flow / flow (m^3)** — cumulative forward-flow counter; consumption = difference of consecutive readings; should be monotonically non-decreasing (a decrease is the “counter rollback” anomaly).
- **Back Flow (m^3)** — cumulative reverse-flow counter; nonzero back-flow can indicate pressure issues or tampering.
- **Sleep Times** — count of device sleep events.
- **Communication Abnormality Times** — count of transmission problems per meter; ties to RSRP/RSSI and to transmission gaps.
- **Sampling architecture** — measure hourly, transmit twice daily; use the measurement timestamp (`dataTime`), not transmission time, to order the series.
- **Commissioning period** — leading run of zero-flow readings before a meter is plumbed in; trimmed per meter at the first-flow timestamp.

A.5 Anomaly Categories (framework)

- **Point** — a single reading anomalous on its own (spike, drop, counter rollback). Often catchable by thresholds.
- **Contextual** — a normal value that is anomalous *for its context* (night-time burst, abnormal profile). Needs context features.
- **Collective** — a sequence anomalous as a group though each point is normal (stuck value, replay, time shift). Needs windowed features.

Implication: the feature matrix must include raw, context, and windowed features to cover all three categories.